Check points of three chapter from java
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1:- (a) List six relational operators ?
Answer: >, <, >=, <=,!=,==
(b) Assuming **x** is **1**, show the result of the following Boolean expressions
(x > **0**) true
(x < **0**) false
(x != **0**) true
(x >= **0**) true
(x != **1)** **false**

(c) Can the following conversions involving casting be allowed? Write a test program to verify it

```
boolean b = true;
i = (int)b;
int i = 1;
boolean b = (boolean)i;
```

answer: no we can not cast because boolean has only two value (true and and false)

2:- (a) write an **if** statement that assigns **1** to **x** if **y** is greater than **0**.
answer:
???

(b) Write an **if** statement that increases pay by 3% if **score** is greater than **90**

answer:

```
import java.util.Scanner;

public class Main {
 public static void main(String[]args) {
 Scanner input = new Scanner(System.in);
 System.out.println("enter the value of payment:");
 double pay = input.nextDouble();
 System.out.println("enter the score:");
 int y = input.nextInt();
 if (y > 90) {
 pay +=pay * 0.03;
 System.out.println("payment equals to:" + pay);
       }
    }
}
```

(c) What is wrong in the following code?

```
if radius >= 0
{
area = radius * radius * PI;
System.out.println("The area for the circle of " +
" radius " + radius + " is " + area);
}
```

ansewr: there are two mistake
. there is no any parenthesis for the conditional of if
. there pi is not initialized

3:- (a) Write an **if** statement that increases **pay** by 3% if **score** is greater than **90**, otherwise increases **pay** by 1%.

answer:

```
import java.util.Scanner;

public class Main {
   public static void main(String[]args) {
     Scanner input = new Scanner(System.in);
       System.out.println("enter the value of payment:");
       double pay = input.nextDouble();
       System.out.println("enter the score:");
       int y = input.nextInt();
       if (y > 90) {
          pay +=pay * 0.03;
```

```
        System.out.println("payment equals to:" + pay);
      }else{
        pay +=pay * 0.01;
        System.out.println("payment equals to:" + pay);
      }
   }
}
```

(b) What is the output of the code in (a) and (b) if **number** is **30**? What if **number** is **35**?

```
if (number % 2 == 0)
  System.out.println(number
    + "is even.");

System.out.println(number
    + "is odd");
```

(a)

```
if (number % 2 == 0)
  System.out.println(number
    + "is even.");
else
System.out.println(number
    + "is odd");
```

(b)

answer: (a) 30 is even , 30 is odd and 35 is odd answer: (b) 30 is even and 35 is odd.

4:-

(a):

Suppose **x = 3** and **y = 2**; show the output, if any, of the following code. What is the output if **x = 3** and **y = 4**? What is the output if **x = 2** and **y = 2**? Draw a flowchart of the code.

```
if (x > 2) {
if (y > 2) {
z = x + y;
System.out.println("z is " + z);
  }
}
else
System.out.println("x is " + x);
```

if x = 2 and y = 2 the out put is empty
if x = 3 and y = 4 the out will be : z is = 7
if x = 2 and y = 2 the else will execute and it will be : x is = 2

(b):

Suppose **x = 2** and **y = 3**. Show the output, if any, of the following code. What is
the output if **x = 3** and **y = 2**? What is the output if **x = 3** and **y = 3**?

```
if (x > 2)
 if (y > 2) {
  int z = x + y;
   System.out.println("z is " + z);
   }
  else
System.out.println("x is " + x);
```

answer: if x=2 and y = 3. the output will be : x is 2 because the else statement will execute

if x = 3 and y = 2 the result will be x is 3 because the also elase staement execute.

(c):

What is wrong in the following code?

```
if (score >= 60)
System.out.println("D");
else if (score >= 70)
System.out.println("C");
else if (score >= 80)
System.out.println("B");
else if (score >= 90)
System.out.println("A");
else
System.out.println("F");
```

answer: this code is not from small number to large number therefore if the first statement becomes true so other else-if and else statement s will not be checked every score that will be equal or obove to 60 the out put will be D.

5:-

(a): Which of the following statements are equivalent? Which ones are correctly indented?

```
if (i > 0) if
(j > 0)
x = 0; else
if (k > 0) y = 0;
else z = 0;
```

(a)

```
if (i > 0) {
  if (j > 0)
    x = 0;
  else if (k > 0)
    y = 0;
}
else
  z = 0;
```

(b)

```
if (i > 0)
  if (j > 0)
    x = 0;
  else if (k > 0)
    y = 0;
  else
    z = 0;
```

(c)

```
if (i > 0)
  if (j > 0)
    x = 0;
  else if (k > 0)
    y = 0;
else
  z = 0;
```

(d)

?????????

(b): Rewrite the following statement using a Boolean expression:

```
if (count % 10 == 0)
newLine = true;
else
newLine = false;
```

answer: ??????/

(c): Are the following statements correct? Which one is better?

```
if (age < 16)
  System.out.println
    ("Cannot get a driver's license");
if (age >= 16)
  System.out.println
    ("Can get a driver's license");
```

(a)

```
if (age < 16)
  System.out.println
    ("Cannot get a driver's license");
else
  System.out.println
    ("Can get a driver's license");
```

(b)

Answer: both of these statement are correct and we use either, but B is better if we could have an else statement after every if statement it seems better than having two if statement.

(d): What is the output of the following code if **number** is **14**, **15**, or **30**?

```
if (number % 2 == 0)
  System.out.println
    (number +  " is even");
if (number % 5 == 0)
  System.out.println
    (number +  " is multiple of 5");
```

(a)

```
if (number % 2 == 0)
  System.out.println
    (number +  " is even");
else if (number % 5 == 0)
  System.out.println
    (number +  " is multiple of 5");
```

(b)

answer:

(a)

if the number is 14, output = 14 is even.

If the number is 15, output = 15 is multiple of 5.

if the number is 30, output = 30 is even , 30 is multiple of 5.

(b)

if the number is 14, output = 14 is even.

If the number is 15, output = 15 is multiple of 5.

if the number is 30, output = 30 is even.

6:-

(a): Which of the following is a possible output from invoking **Math.random()**?

323.4, **0.5**, **34**, **1.0**, **0.0**, **0.234**

answer:?????

(b):

a. How do you generate a random integer **i** such that 0 … *i* 6 20?

b. How do you generate a random integer **i** such that 10 … *i* 6 20?

c. How do you generate a random integer **i** such that 10 … *i* … 50?

d. Write an expression that returns **0** or **1** randomly.

?????? what is the random????

7: Are the following two statements equivalent?

```
if (income <= 10000)
  tax = income * 0.1;
else if (income <= 20000)
  tax = 1000 +
    (income - 10000) * 0.15;
```

```
if (income <= 10000)
  tax = income * 0.1;
else if (income > 10000 &&
         income <= 20000)
  tax = 1000 +
    (income - 10000) * 0.15;
```

??????????

8:- (a): Assuming that **x** is **1**, show the result of the following Boolean expressions:

(**true**) && (**3** > **4**)------->false
!(x > **0**) && (x > **0**)------->false
(x > **0**) || (x < **0**)--------->true
(x != **0**) || (x == **0**)------>true
(x >= **0**) || (x < **0**)------->true
(x != **1**) == !(x == **1**)------→true

(b): Write a Boolean expression that evaluates to **true** if a number stored in variable **num** is between **1** and **100**. (b) Write a Boolean expression that evaluates to **true** if a number stored in variable **num** is between **1** and **100** or the number is negative.

Answer:

(a) boolean number = (num>1 && num<100);
(b) boolean number = (num>1 && num<100) || (num<0);

(c): (a) Write a Boolean expression for $|x - 5| < 4.5$. (b) Write a Boolean expression for $|x - 5| > 4.5$

answer:

(a) [(x – 5)<4.5 && (x – 5)>4.5]
(b) [(x – 5)>4.5 && (x – 5)<4.5]

(d) Assume **x** and **y** are **int** type. Which of the following are legal Java expressions?

x > y > **0**------>**illegal**
x = y && y----->illegal
x /= y------->illegal
x or y------->illegal
x and y-------->illegal
(x != **0**) || (x = **0**)-----→legal

(e): Are the following two expressions the same?

(a) x % **2** == **0** && x % **3** == **0**
(b) x % **6** == **0**
answer: yes because if a number able to be divided by 2 and 3 in the same time so it also divide able by 6.

(f): What is the value of the expression **x >= 50 && x <= 100** if **x** is **45**, **67**, or **101**?
answer: if x is 45 the value is false, if x is 67 the value is true and if x is 101 the value is false.
Answer:????????
(g) Write a Boolean expression that evaluates to **true** if **age** is greater than **13** and less than **18**.
[(age >13) && (age<18)]
(h) Write a Boolean expression that evaluates to **true** if **weight** is greater than **50** pounds or height is greater than **60** inches.
Answer: [(weight > 50) || (height > 60)]
(I) Write a Boolean expression that evaluates to **true** if either **weight** is greater than **50** pounds or height is greater than **60** inches, but not both.
(weight > 50 ^ height > 60).

9: How many days in the February of a leap year? Which of the following is a leap year? 500, 1000, 2000, 2016, and 2020?
answer:????????

10:- What happens if you enter an integer as **05**?
answer: if we enter an integer as 05 the output will be 5 and zero the before of the integer will ignore.

11:- (a): What data types are required for a **switch** variable? If the keyword **break** is not used after a case is processed, what is the next statement to be executed? Can you convert a **switch** statement to an equivalent **if** statement, or vice versa? What are the advantages of using a **switch** statement?
Answer: for switch variables the data type like String , integer , char , short , byte can be used.

If the keyword break is not used when it is needed it may cause error because if the case matches so the code will be executed till end of the switch statement is reached, it means that all case will executed till end.

In most cases we can convert them one to another but not at all for example if an interval needed statement is given to if it is difficult to convert it to a switch statement, so both has their advantages in special cases and a switch statement can make a long If statement code to a simple one and switch statement is used to simplify coding for multiple condition.

(b): What is **y** after the following **switch** statement is executed? Rewrite the code using an **if-else** statement.

```
x = 3; y = 3;
switch (x + 3) {
case 6: y = 1;
default: y += 1;
}
```

????????/

(c): What is **x** after the following **if-else** statement is executed? Use a **switch** statement to rewrite it and draw the flowchart for the new **switch** statement.

```
int x = 1, a = 3;
if (a == 1)
  x += 5;
else if (a == 2)
  x += 10;
else if (a == 3)
  x += 16;
else if (a == 4)
x += 34;
```

answer: x will be 17 after the executed.

```
Int x=1; int a = 3;
switch(a) {
case 1 : x +=5; break;
case 2 : x +=10; break;
case 3 : x +=16; break;
case 4 : x +=34; break;
}
```

(d): Write a **switch** statement that displays Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, if **day** is **0**, **1**, **2**, **3**, **4**, **5**, **6**, respectively.

Answer:

```
switch(day of weak) {
case 0 : System.out.print(“saturday”) ; break;
case 1 : System.out.print(“sunday”); break;
case 2 : System.out.print(“monday”); break;
case 3 : System.out.print(“tuesday”); break;
```

case 4 : System.out.print(“widnesday”); break;
case 5 : System.out.print(“tursday”); btreak;
case 6 : System.out.print(“friday”); break;
}
?????

12:- Rewrite the following **if** statements using the conditional operator.

```
if (ages >= 16)
  ticketPrice = 20;
else
  ticketPrice = 10;
```

Answer: int age = input.nextLine();
double ticketprice = (age >= 16)? 20 : 10 ;